

|                   |                                |                                                                                                                                                                                                                                                                                                                                                     |
|-------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   |                                | <b>Stream</b><br><code>.filter(...)</code> // reduce data<br><code>.distinct()</code> // remove duplicates<br><code>.map(...)</code> // transform<br><code>.flatMap(...)</code> // flatten if needed<br><code>.sorted(...)</code> // expensive → late<br><code>.peek(...)</code> // debug only<br><code>.collect(...);</code> // terminal operation |
| <b>Operation</b>  | <b>Use Case</b>                | <b>One-Line Code Example</b>                                                                                                                                                                                                                                                                                                                        |
| <b>filter()</b>   | boolean checks                 | <code>.filter(x -&gt; x &gt; 10)</code>                                                                                                                                                                                                                                                                                                             |
| <b>filter()</b>   | null checks                    | <code>.filter(Objects::nonNull)</code>                                                                                                                                                                                                                                                                                                              |
| <b>filter()</b>   | comparisons                    | <code>.filter(e -&gt; e.getSalary() &gt;= 50000)</code>                                                                                                                                                                                                                                                                                             |
| <b>filter()</b>   | method calls returning boolean | <code>.filter(User::isActive)</code>                                                                                                                                                                                                                                                                                                                |
| -                 |                                |                                                                                                                                                                                                                                                                                                                                                     |
| <b>distinct()</b> | rely on equals() / hashCode()  | <code>.distinct()</code>                                                                                                                                                                                                                                                                                                                            |
| <b>distinct()</b> | immutable objects              | <code>.map(String::toUpperCase).distinct()</code>                                                                                                                                                                                                                                                                                                   |
| <b>map()</b>      | transform objects              | <code>.map(Employee::getName)</code>                                                                                                                                                                                                                                                                                                                |
| <b>map()</b>      | create new objects             | <code>.map(e -&gt; new EmployeeDTO(e.getId(), e.getName()))</code>                                                                                                                                                                                                                                                                                  |
| <b>map()</b>      | call pure functions            | <code>.map(Math::abs)</code>                                                                                                                                                                                                                                                                                                                        |

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| <b>flatMap()</b> | return Stream       | .flatMap(s -> Arrays.stream(s.split(",")))                                          |
| <b>flatMap()</b> | split data          | .flatMap(sentence -> Arrays.stream(sentence.split(" ")))                            |
| <b>flatMap()</b> | expand collections  | .flatMap(order -> order.getItems().stream())                                        |
| -                |                     |                                                                                     |
| <b>sorted()</b>  | natural ordering    | .sorted()                                                                           |
| <b>sorted()</b>  | custom comparator   | .sorted(Comparator.comparing(Employee::getSalary))                                  |
| <b>sorted()</b>  | multi-field compare | .sorted(Comparator.comparing(Employee::getDept).thenComparing(Employee::getSalary)) |
| <b>peek()</b>    | logging             | .peek(e -> System.out.println(e))                                                   |
| <b>peek()</b>    | debugging           | .peek(e -> log.debug("Processing: {}", e))                                          |
| <b>peek()</b>    | metrics             | .peek(e -> counter.incrementAndGet())                                               |
| -                |                     |                                                                                     |
| <b>collect()</b> | accumulate data     | .collect(Collectors.toList())                                                       |
| <b>collect()</b> | grouping            | .collect(Collectors.groupingBy(Employee::getDepartment))                            |
| <b>collect()</b> | reduction           | .collect(Collectors.summingInt(Employee::getSalary))                                |
| <b>collect()</b> | custom collectors   | .collect(Collector.of(ArrayList::new, List::add, (a,b)->{a.addAll(b);return a;}))   |